Chirag Taneja

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Education

Thapar Institute of Engineering and Technology — Patiala, India
BE in Robotics and Artificial Intelligence — *2023 – 2027*

Experience

Research Intern — June 2025 – August 2025
The University of Queensland — *Brisbane, Australia*

- Developed AI explainability techniques for deepfake detection systems, enhancing transparency in social media content moderation pipelines
- Built interpretable ML models to generate human-readable explanations for deepfake classification decisions under Dr. Mashuda Glencross and Dr. Suresh Raikwar

Projects

Reisearch: Multi-Agent Deep Research System | *Python, LangGraph, Groq, Tavily*

- Architected a supervisor-worker multi-agent system using LangGraph subgraphs that decomposes vague queries into scoped research briefs, delegates parallel sub-tasks to specialized agents, and synthesizes comprehensive markdown reports
- Implemented a **Scoping Agent** with iterative clarification loops and a **Supervisor Agent** with a thinking tool that dynamically parallelizes research briefs into sub-tasks via LangGraph state transitions
- Added memory checkpointing for workflow persistence, exponential backoff retry systems for API resilience, and LangSmith evals for deterministic agent benchmarking
- GitHub: ctxnn/deepresearch-agent

Tri-modal AI Agent (Chat + RAG + Code) | *Python, LangGraph, LangChain, OpenRouter*

- Built a LangGraph state-machine agent with conditional routing across chat, RAG, and code generation modes with Human-in-the-Loop approval gates for safe autonomous execution
- Integrated custom vector embeddings for document retrieval and sandboxed subprocess workspaces for code agent isolation, with conditional edges routing user intent to appropriate modes
- Engineered memory persistence via `InMemorySaver` and a tri-modal intent router using LangGraph nodes and conditional edges
- GitHub: ctxnn/CKC-agent

GPT-2 Language Model | *Python, PyTorch, Hugging Face, Gradio*

- Implemented and pretrained a **124M-parameter GPT-2 from random initialization** on **9.99B FineWeb-Edu tokens** using an H100; achieved **3.03 validation loss**, **20.71 perplexity**, and **30.03% HellaSwag accuracy**
- Built a verified 100-shard data pipeline and resumable S3 checkpoints preserving optimizer, dataloader, and RNG state across interrupted jobs; converted the model to Hugging Face safetensors and deployed a public Gradio demo

Positions of Responsibility

General Secretary | Mecivions Research Society — January 2026 – Present
Thapar Institute of Engineering and Technology — *Patiala, India*

Executive Member | Rotaract Club — August 2024 – July 2025
Thapar Institute of Engineering and Technology — *Patiala, India*

Technical Skills

Agentic AI & Orchestration: LangGraph, LangChain, Multi-Agent Systems, Supervisor-Worker Patterns, State Management, Tool Calling
Languages: Python, C++, C
Machine Learning: PyTorch, TensorFlow, Scikit Learn
LLMs & Inference: Groq, OpenRouter, Hugging Face
Deep Learning: Transformer, MoE, LoRA, Diffusion Models, GAN, VAE, Autoencoder
Retrieval & Search: RAG, Vector Embeddings, Tavily, Web Search Integration
Developer Tools: Git, VS Code, PyCharm, Jupyter, CLI, Linux, LangSmith